

Panorama Generation from Overlapping Photos:

An Advanced Modular Pipeline

C05430 - Image Processing
(Project P09, Group G16)

Group Members

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Problem & Motivation

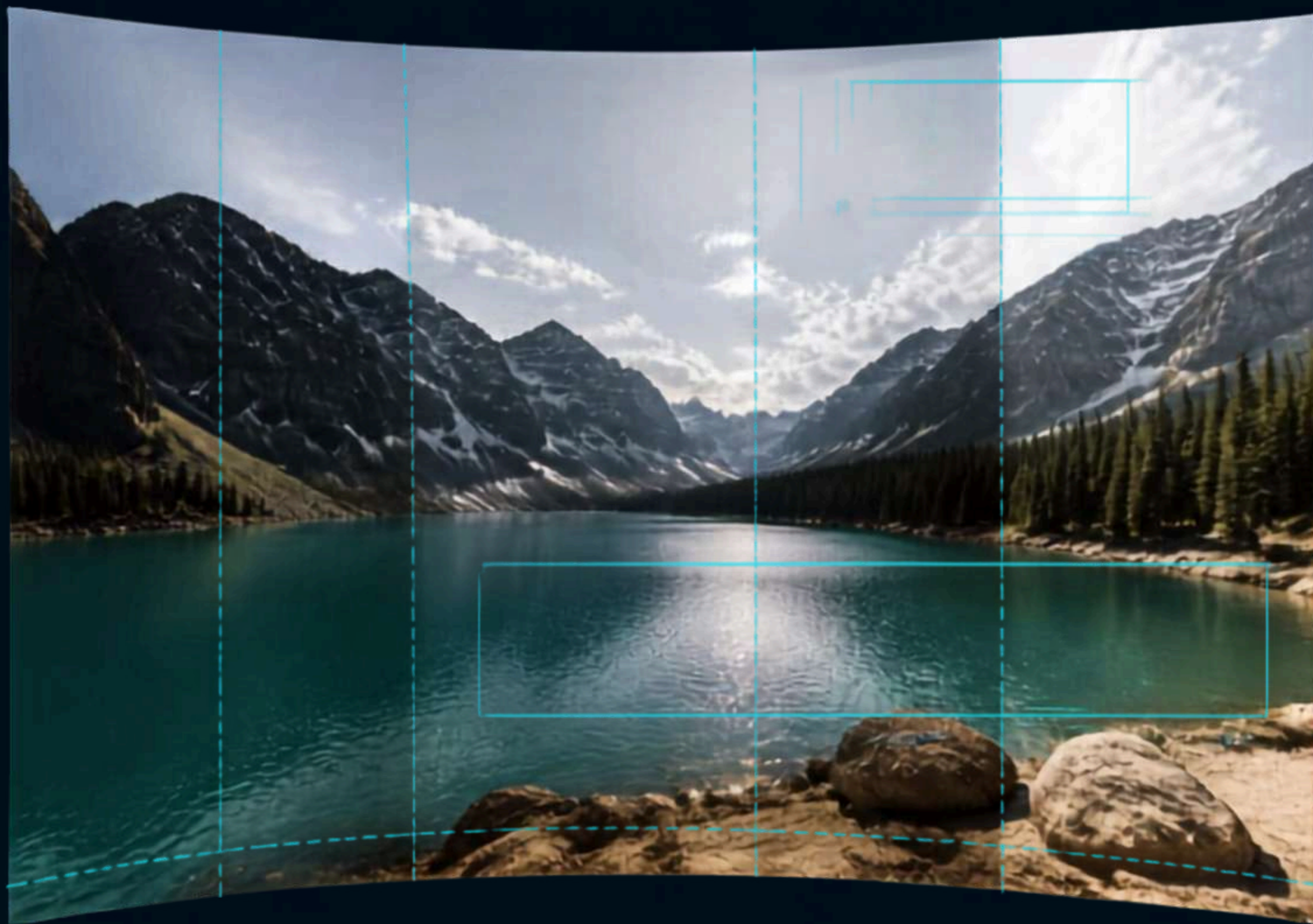
Build a **seamless**, wide field-of-view panorama from overlapping photos.

KEY CHALLENGES

Exposure variations causing visible seams

Geometric stretching distortions

Cumulative drift in multi-image stitching sequences



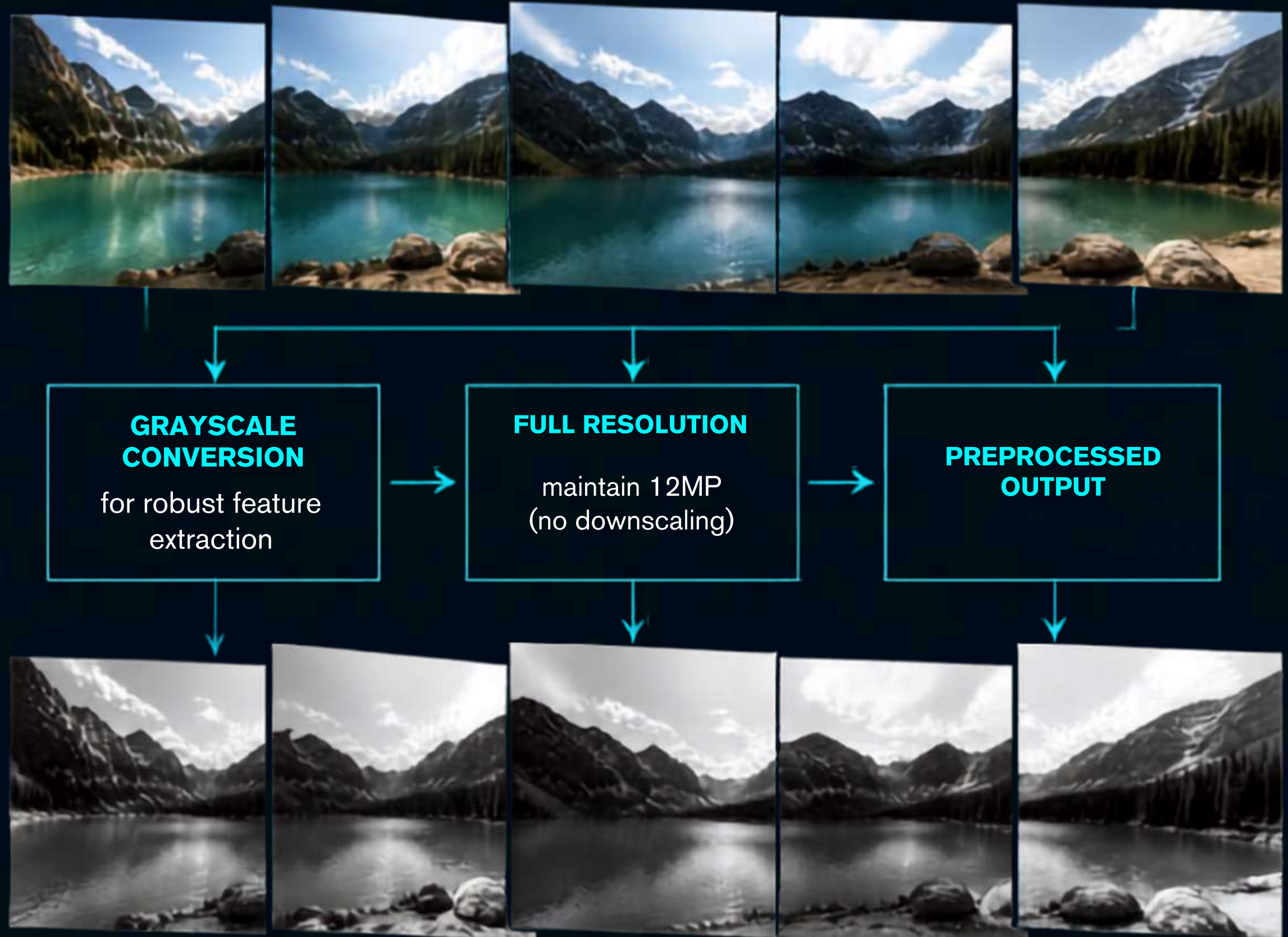
Dataset & Preprocessing

Custom smartphone dataset captured outdoors under natural lighting

Converted images to grayscale for robust feature extraction

Maintained full **12MP** resolution (no downscaling) for accuracy

INPUT: OVERLAPPING PHOTOS

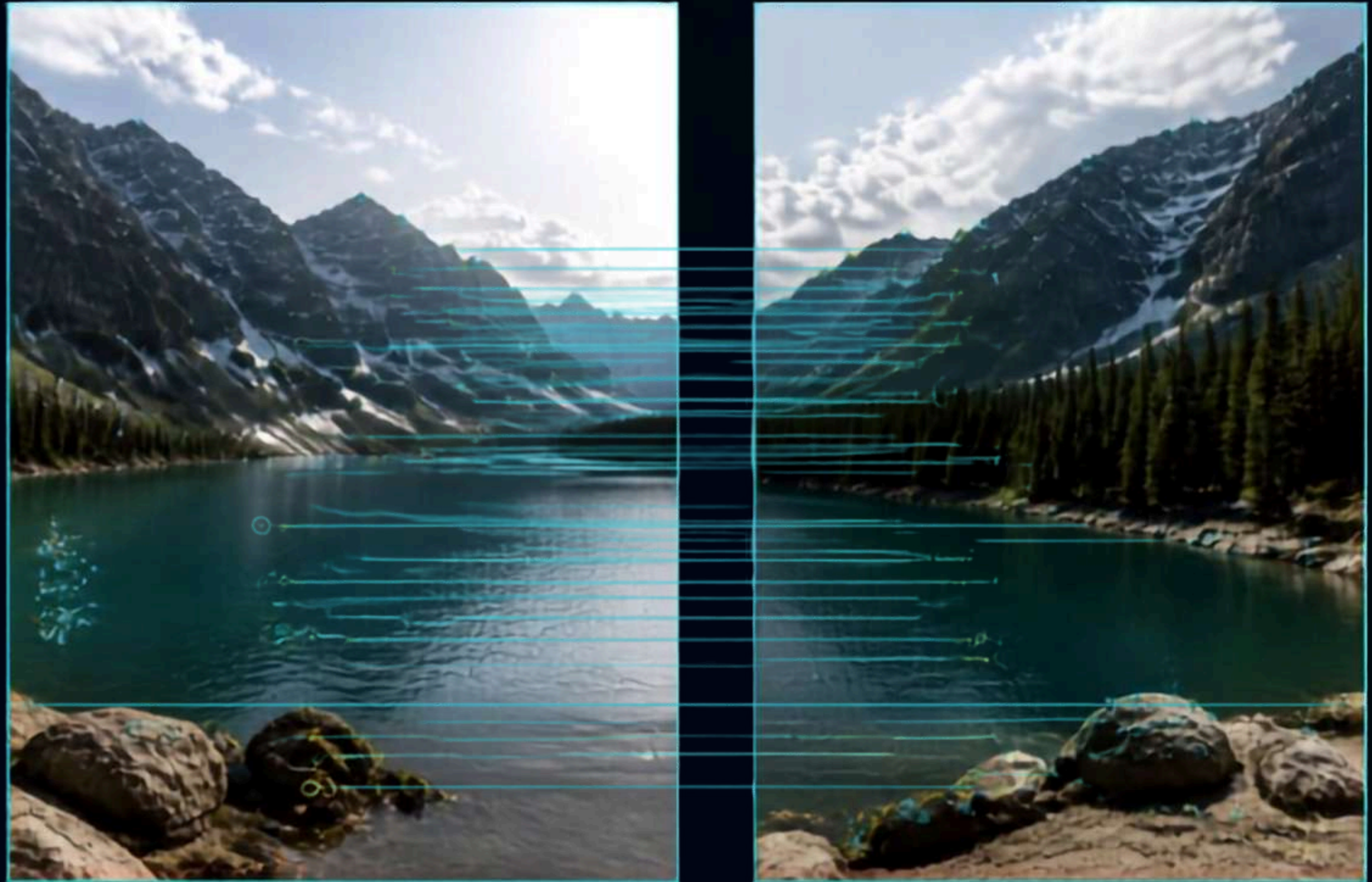


Feature Detection & Matching

Replaced baseline ORB **SIFT** for
scale and rotation invariance

Used **FLANN** with KD-Trees for
fast approximate nearest
neighbor search

Applied Lowe's **Ratio Test**
(threshold 0.70) to filter
ambiguous matches



Homography Estimation & Smart Affine Fallback

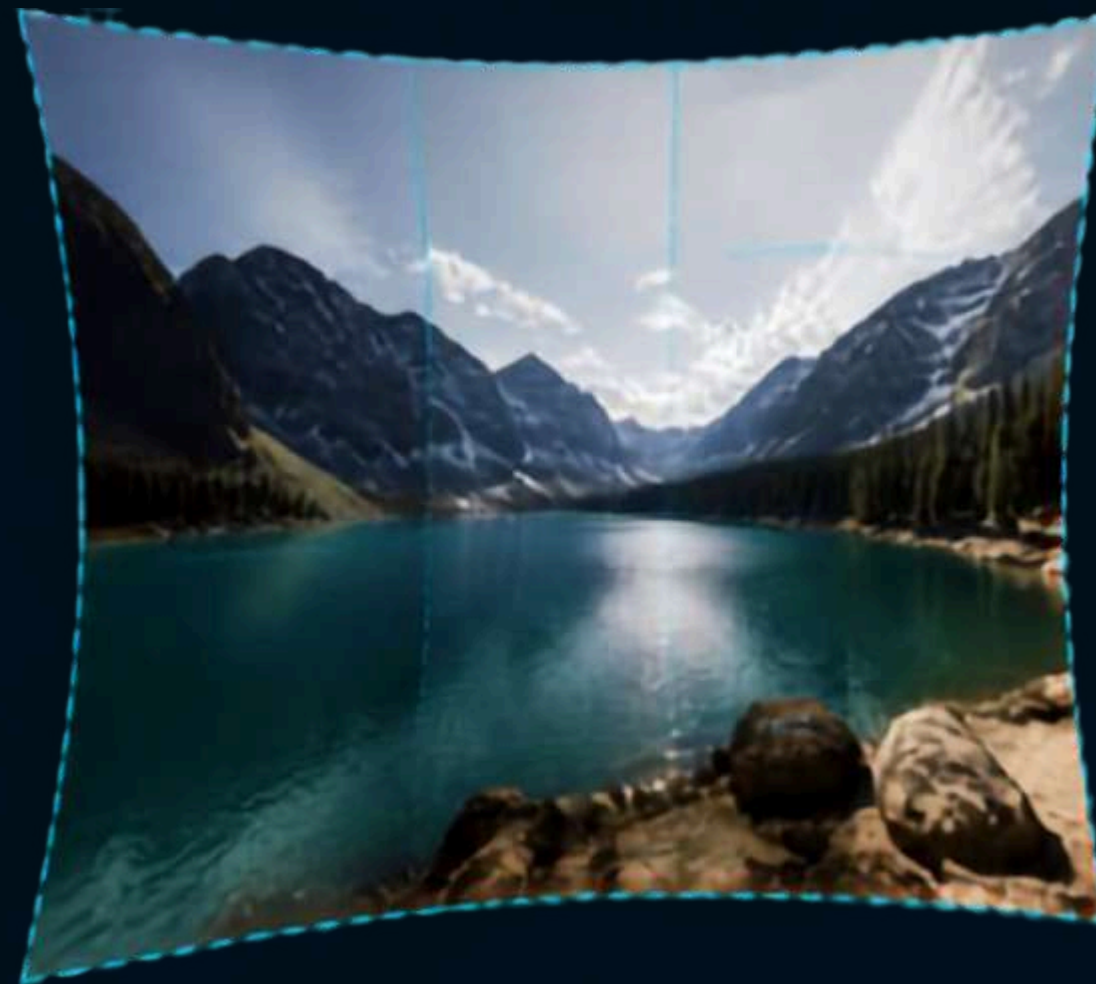
Used **RANSAC** (5.0 px threshold) to estimate 3x3 projective homography matrix

Smart Fallback

If homography causes $>1.3x$ stretch, fallback to 2D affine transform

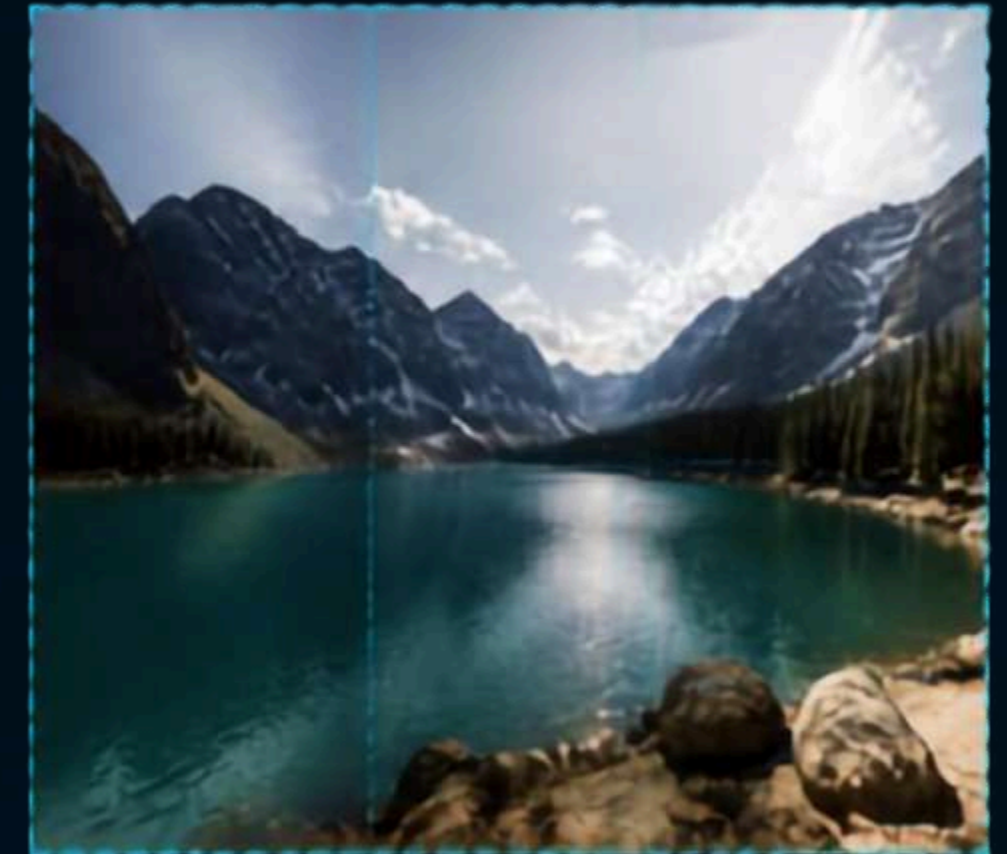
Prevents infinite perspective distortion in challenging cases

HOMOGRAPHY RESULT



Excessive stretch ($>1.3x$)
leads to perspective distortion

AFFINE RESULT

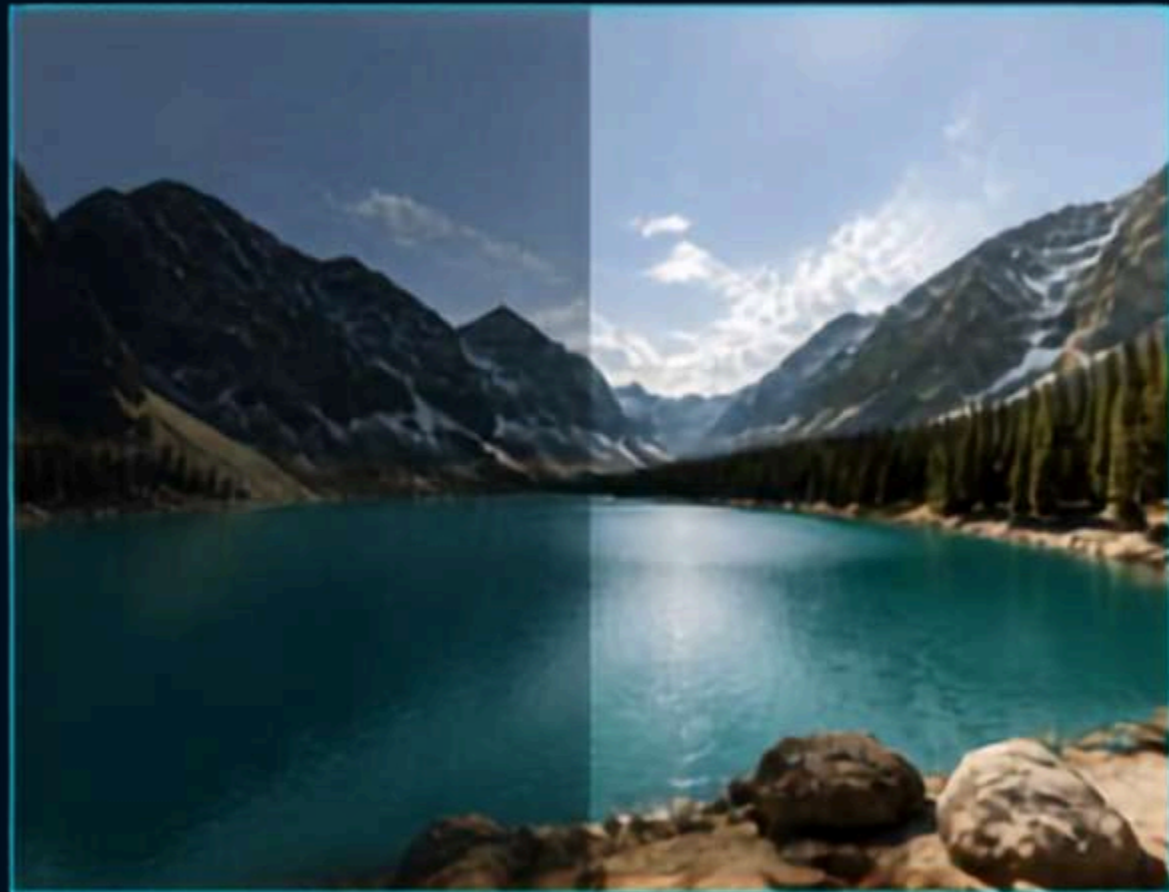


Affine fallback
maintains stable geometry

Advanced Image Blending

Seamless panoramas with superior visual quality

BASELINE OVERWRITE



Harsh visible seams
from exposure mismatch

FEATHERING



Smoother transitions
with slight blurring

MULTI-BAND BLENDING



No visible seams while
preserving detail and texture

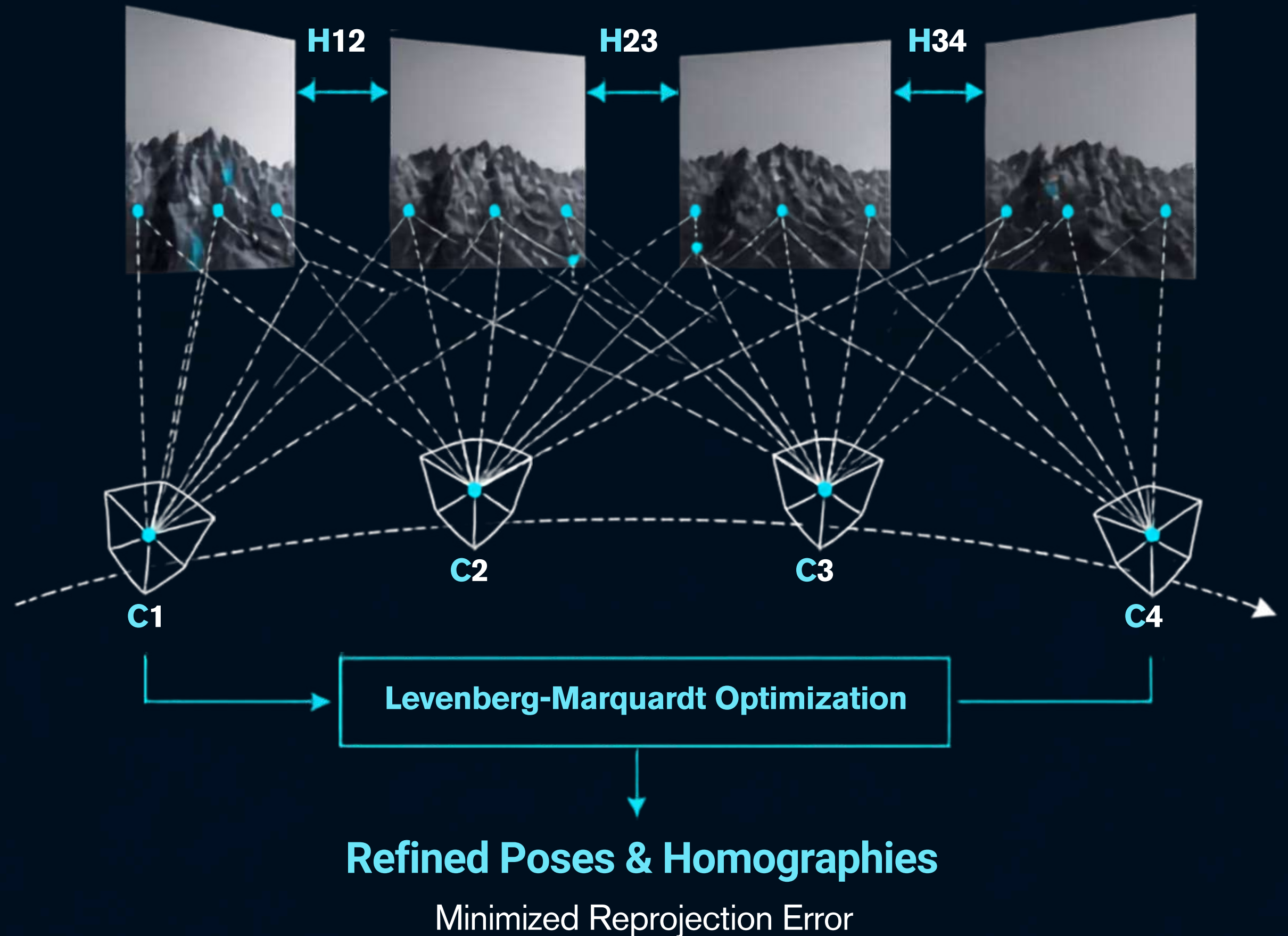
Global Pose Refinement

Bundle Adjustment reduces **drift** across the panorama.

Simultaneously refines all camera poses and pairwise homographies

Optimized with Levenberg-Marquardt algorithm

Minimizes total reprojection error across images



Quantitative Results

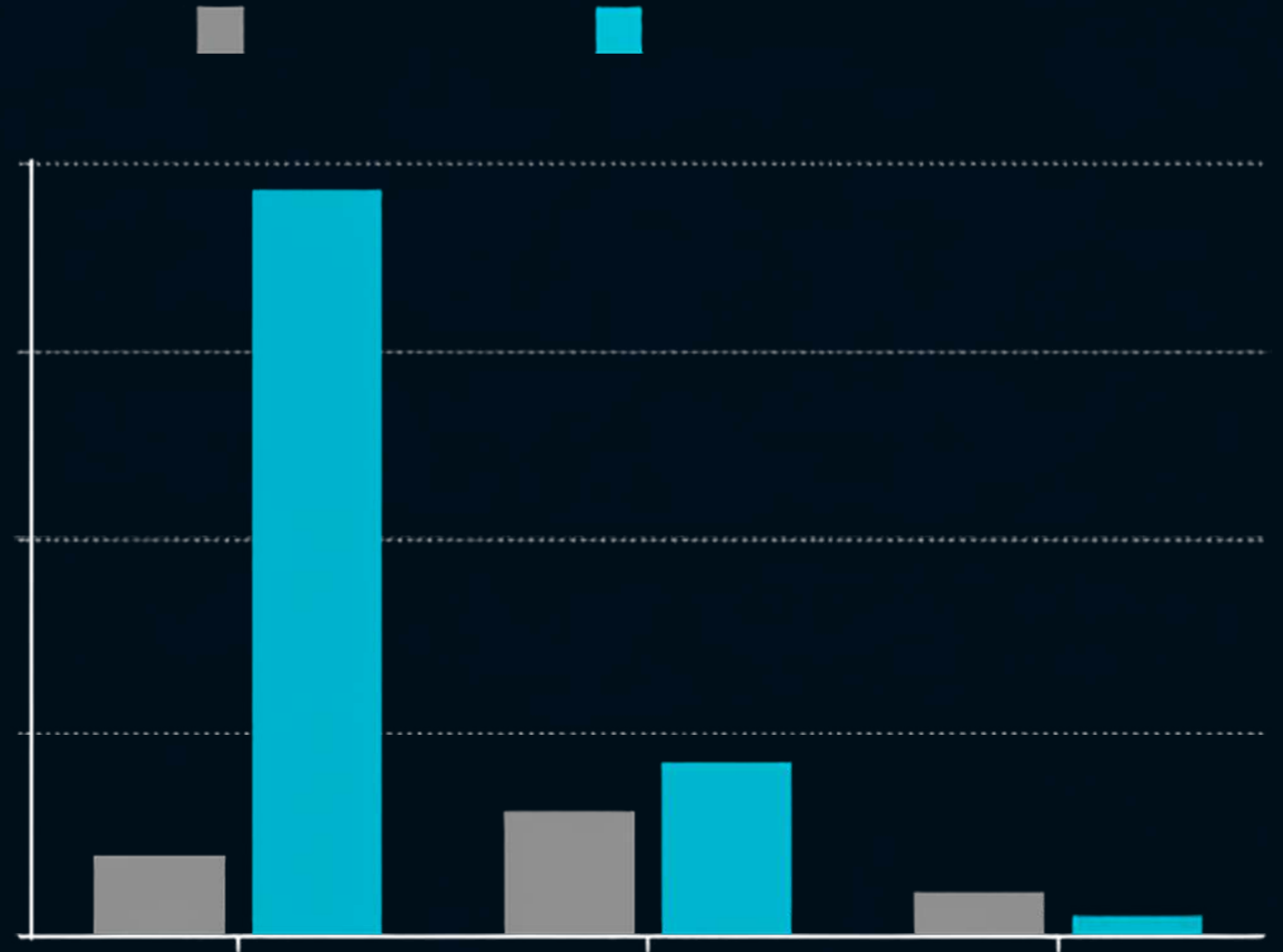
Our advanced pipeline delivers major gains across all key metrics.

384 matches achieved with the final SIFT pipeline

76.56% inlier ratio in the final result

0.54 px RMSE with 3.8× improvement

Achieved **sub-pixel accuracy** in feature alignment.



Failure Cases & Limitations

GHOSTING ARTIFACTS

Moving objects leave duplicate or translucent trails.

PARALLAX ERRORS

Handheld capture and depth variation cause misalignments and warping.

LIMITED ROBUSTNESS

Extreme lighting, repetitive textures, and low overlap reduce reliability.



Conclusion & Demo

Built a modular, production-grade
panorama stitching pipeline

Delivered **visually flawless**
panoramas with multi-band
blending and bundle adjustment

Next: Live demo

See the pipeline in action
and the output results

